Exercise 1.4

1. It is viable 2. It is not viable 3. It is viable
4. A = 27.82 tonnes B = 98.91 tonnes 5. 181.62, 84.32
6. 34.16, 17.31, 7. 42 and 78

Exercise 1.5

1	2	3	4	5	6	7	8	9	10	11	12	13
(b)	(d)	(b)	(b)	(c)	(c)	(c)	(c)	(c)	(d)	(c)	(b)	(a)
14	15	16	17	18	19	20	21	22	23	24	25	
(c)	(c)	(c)	(b)	(a)	(d)	(b)	(b)	(d)	(d)	(b)	(a)	

Miscellaneous Problems

1. $x = 3. x = -1$ 2. 0 6. $\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$
7. $\frac{1}{5}\begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$ 8. $x = 1, y = 2, z = 3$ 9. $x = 20, y = 30, z = 50$
10. ₹1200 Crores, ₹1600 Crores

2. ALGEBRA

Exercise 2.1

1. $\frac{13}{x-2} - \frac{10}{x-1}$ 2. $\frac{3}{x-2} + \frac{1}{x+1}$
3. $\frac{1}{9(x-1)} - \frac{1}{9(x+2)} - \frac{1}{3(x+2)^2}$ 4. $\frac{1}{2(x-1)} - \frac{1}{2(x+1)}$
5. $\frac{4}{9(x-1)} - \frac{4}{9(x+2)} - \frac{1}{3(x-1)^2}$ 6. $\frac{2}{x-2} + \frac{3}{(x-2)^2} - \frac{9}{(x-2)^3}$
7. $\frac{-7}{2x} + \frac{1}{x^2} + \frac{9}{2(x+2)}$ 8. $\frac{1}{5(x+2)} + \frac{4(x-2)}{5(x^2+1)}$
9. $\frac{3}{16(x-1)} - \frac{3}{16(x+3)} + \frac{1}{4(x+3)^2}$ 10. $\frac{1-x}{5(x^2+4)} + \frac{1}{5(x+1)}$

Exercise 2.2

1. 64 2. 10 3. 3 4. 336 5. 85

Exercise 2.3

1. 6 2. 14400 3. 1680 4. $\frac{13!}{4!\,3!\,2!\,2!}$
5. (a) $\frac{7!}{2}$ (b) 7! 6. Rank of the word CHAT = 9.

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